Machine Learning Lectures

Responsible ML

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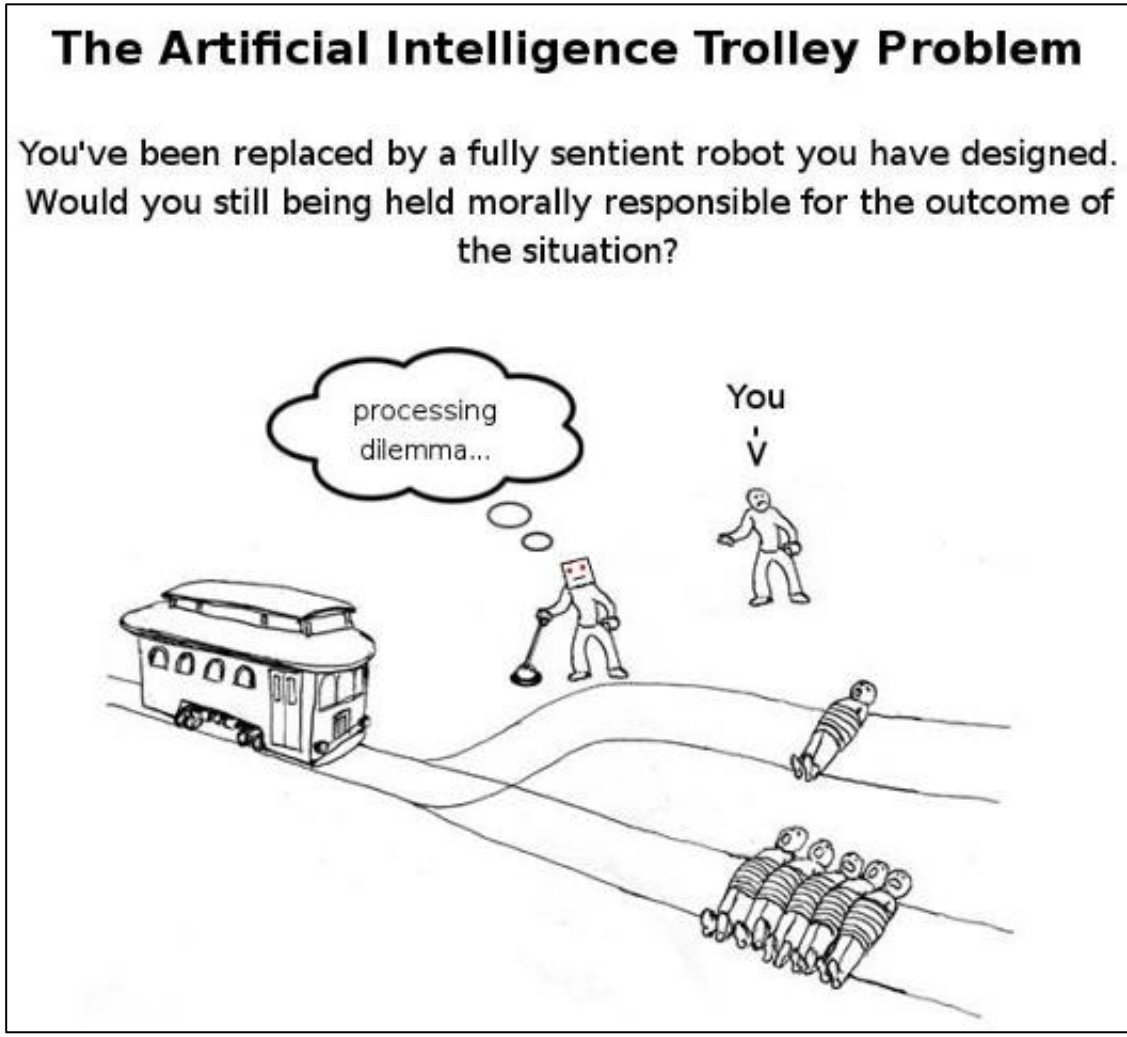
What is Responsible ML?

Responsible ML ensures machine learning models are ethical, fair, transparent, stable, and accountable.

Why Responsible ML Matters?

It protects users, aligns with regulations, and builds trust in AI.



Agenda

1. Introduction to Responsible ML
2. Common Pitfalls and Stability in ML
3. Four Core Principles: Ethics, Transparency, Reliability, Accountability
4. Technical Foundations: Preprocessing, Outlier Detection, Interpretability
5. Bias Mitigation, Privacy, and Security
6. Advanced Topics: Conformal Predictions and Causal Inference
7. Tools and Resources
8. Organizational Best Practices
9. Summary

Why Responsible ML Matters

- Ensures fairness, transparency, reliability, and accountability.
- Builds trust with users and stakeholders.
- Aligns with societal and legal expectations.
- Examples of failure:
  - COMPAS bias in criminal justice (2022) [1].
  - Mortgage-approval algorithms in use in the U.S. were shown to be more likely to reject non-white applicants (2021report by *The Markup*) [2]

[1] **Equal Confusion Fairness: Measuring Group-Based Disparities in Automated Decision Systems:**
https://ieeexplore.ieee.org/document/10029385
[2] **The Secret Bias Hidden in Mortgage-Approval Algorithms**
https://themarkup.org/denied/2021/08/25/the-secret-bias-hidden-in-mortgage-approval-algorithms

Common Pitfalls in ML

- Bias: Historical, sampling, or algorithmic.
- Black-box models: Lack of interpretability.
- Instability: Data drift, real-world changes.
- Security vulnerabilities: Adversarial attacks.

The Concept of Stability in ML

Stability = consistent performance under various conditions

Key aspects:

1. **Data Stability**: Handles distribution shifts.
2. **Hyperparameter Stability**: Robust to small changes.
3. **Operational Stability**: Adapts to real-world changes.

Essential for real-world deployment

Instability Example

- Financial algorithms during market shifts.
  - Overfit to bullish conditions.
  - Failed during sudden downturns.
- Consequences: Financial losses and lost trust.

Principles of Responsible ML

- Ethics
- Transparency
- Reliability & Stability
- Accountability

Principle #1 – Ethics

- **Core Values**: Minimize harm, promote fairness, and respect individual autonomy.
- **Proactive Actions**: Regularly audit models for bias and ensure equitable outcomes.
- **Real-World Example**: Facial recognition systems often perform poorly on darker skin tones—highlighting the need for diverse datasets and fairness checks.

Principle #2 – Transparency

- **Black box** models erode trust and hinder accountability..
- **Make AI decisions understandable**: Utilize SHAP, LIME, and other techniques.
- **Structured Documentation:** Adopt Model Cards to clearly communicate model purpose, performance, and limitations. (https://modelcards.withgoogle.com/about)
- **Outcome:** Transparency builds confidence, ensures ethical use, and strengthens accountability in AI systems.

Principle #3 – Reliability & Stability

- **Test models across diverse scenarios** on varied datasets and real-world scenarios to uncover vulnerabilities.
- **Continuous Monitoring:** Track data drift, concept drift, and evolving user needs.
- **Long-Term Focus:** Design systems to adapt and remain effective over time.

Principle #4 – Accountability

- **Ownership:** Define who is responsible for model decisions, performance, and errors at every stage.
- **Legal Frameworks:** Align with evolving regulations and ethical guidelines to ensure compliance.
- **Example:** Accountability in autonomous vehicles (who is liable in case of accidents) and AI-assisted medical diagnoses (ensuring the safety of patients).
- **Monitoring:** Implement systems for regular audits and transparent reporting.

Technical Foundations – Data Preprocessing

- **The Principle:**
  - **"Garbage in, garbage out"** – model quality depends on data quality.
  - Data errors, imbalances, and bias affect predictions directly.

- **Critical Steps**:
  - **Data Cleaning:** Detect and correct errors, duplicates, and inconsistencies.
  - **Data Validation:** Ensure data integrity, format and logical consistency.
  - **Data Balancing:** Handle imbalanced datasets to prevent model bias.
- **Ethical Data Handling:** Preserve diversity for fairness in predictions. Avoid discarding minority or edge-case data points.

Outlier & Anomaly Analysis

Key Techniques: Isolation Forest, DBSCAN, Statistical Methods

Approach with Caution:

- Avoid blind removal of outliers from minority or vulnerable groups.
- Understand the context before discarding anomalous data.

Why It Matters:

- Improves **fairness** by preserving critical minority data.
- Enhances **stability** by addressing genuine anomalies without overfitting.

Interpretability Methods in ML

Why Interpretability Matters:

- Builds **trust** by explaining decisions clearly.
- Detects and mitigates **biases or errors** in models.
- Aids in debugging by identifying **unstable features** or data issues.
- Ensures **accountability** in critical domains like healthcare and finance.

Approaches:

- **Model-Agnostic Methods**: Work across any model (e.g., SHAP, LIME).
- **Model-Specific Methods**: Tailored to specific architectures (e.g., saliency maps for neural networks).

Fairness in ML: Recognizing and Mitigating Bias

Recognizing Bias:

- **Historical Bias:** Data reflects past societal inequities.
- **Sampling Bias:** Skewed representation in data collection.
- **Algorithmic Bias:** Amplification of existing biases during training.

Fairness Metrics *:

- **Disparate Impact:** Measures how outcomes vary across groups.
- **Equal Opportunity:** Ensures similar true positive rates across groups.

Bias Mitigation Strategies:

- **Data Rebalancing:** oversampling or undersampling underrepresented groups.
- **Adversarial Debiasing**: Adversarial networks to reduce bias during training.
- **Threshold Adjustments**: Adjust decision thresholds to equalize outcomes across groups.

* https://superwise.ai/blog/gentle-introduction-ml-fairness-metrics/

Privacy Considerations in Machine Learning

Key Privacy Techniques:

- Differential Privacy:
  - Adds controlled noise to protect individual records while maintaining data utility.
  - Example: Apple uses differential privacy in its data collection to ensure individual anonymity.
- Federated Learning:
  - Keeps data localized on devices; only aggregated updates are shared.
  - Example: Google uses federated learning in Gboard to improve typing predictions without storing raw user data centrally.

Compliance with regulations: GDPR, CCPA, HIPAA

Reinforcing user trust and legal adherence

Security & Adversarial Robustness

Threats:

- **Evasion Attacks:** Subtle manipulations in input data causing incorrect predictions.
- **Poisoning Attacks:** Corrupting training data to compromise model integrity.

Defensive Measures:

- **Adversarial Training:** Enhance model resilience by training with adversarial examples.
- **Gradient Masking:** Reduce gradient-based attack vectors (beware of over-reliance).

Why It Matters:

- Strong defenses correlate with improved overall stability and robustness in real-world applications.

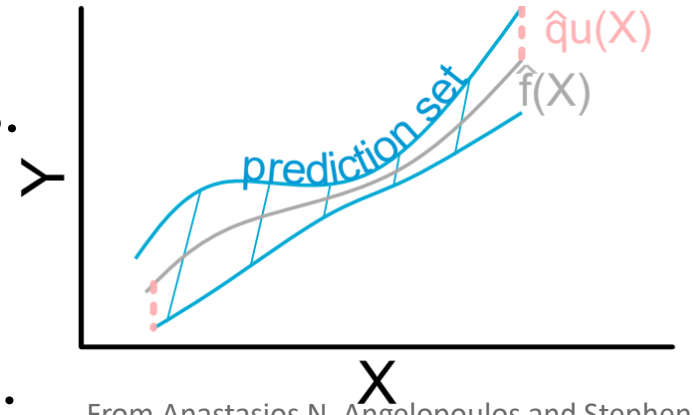
Conformal Predictions: Robust Uncertainty Quantification

Provide Prediction Intervals with Guarantees

- Deliver ranges of outcomes with confidence levels.
- Ensure reliable error bounds for any prediction.

Quantify Uncertainty

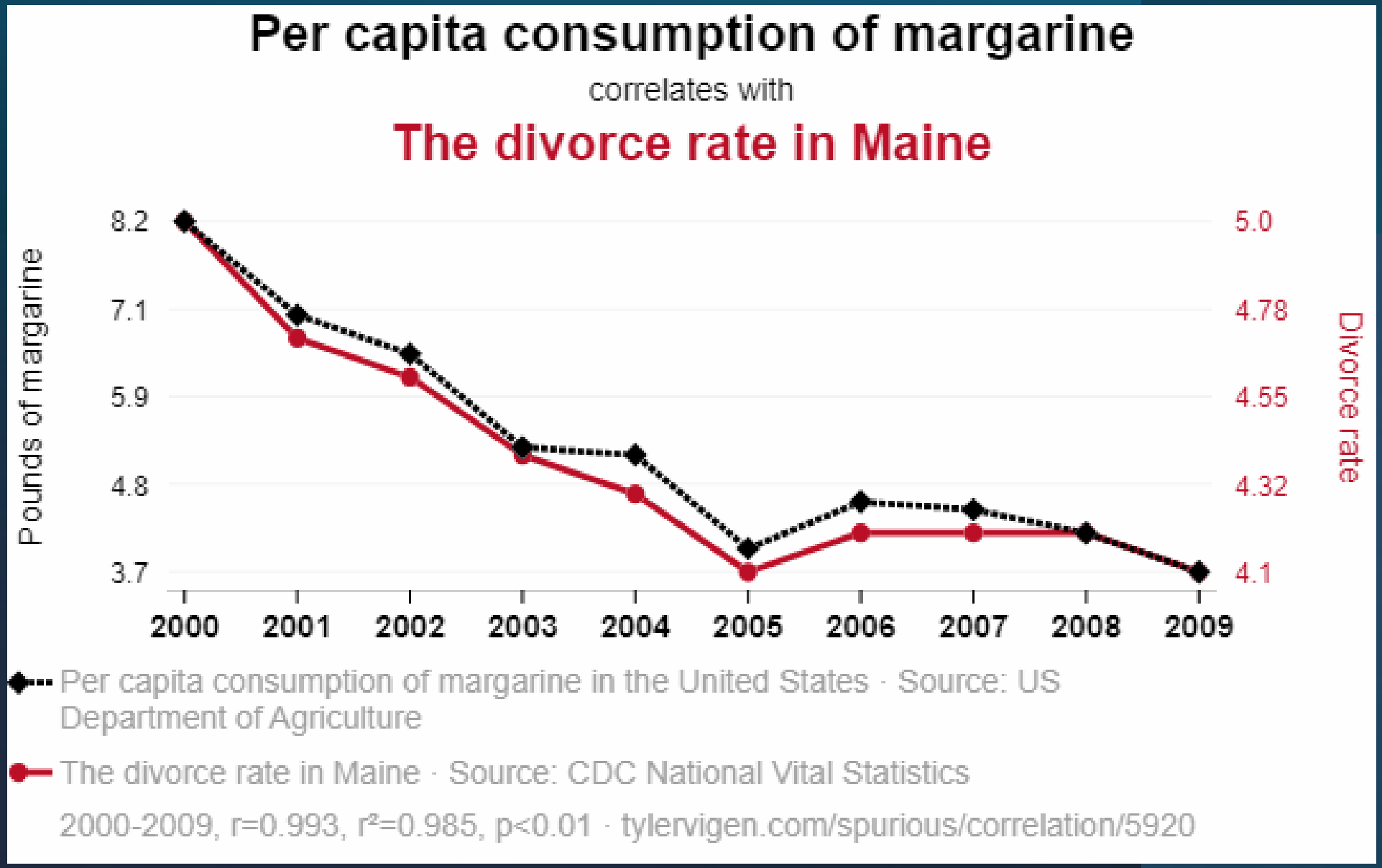
- Avoid overconfident or misleading point estimates.
- Highlight areas of model ambiguity.



From Anastasios N. Angelopoulos and Stephen Bates: A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification

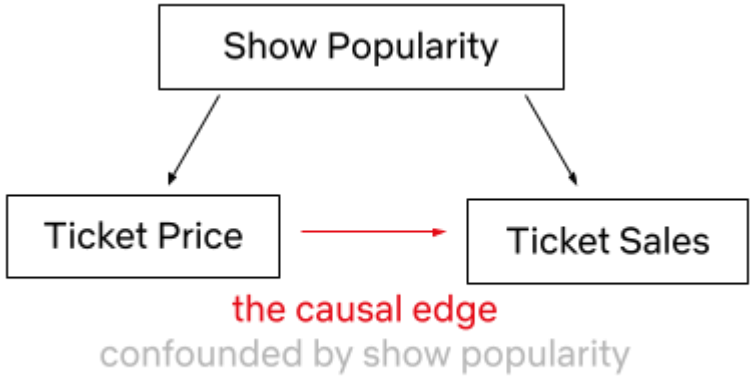
Critical Applications

- Healthcare: Predict disease risk with actionable ranges.
- Finance: Estimate stock volatility with confidence bands.



Spurious correlations

Causal ML (Causal Inference)



https://netflixtechblog.com/causal-machine-learning-for-creative-insights-4b0ce22a8a96

Key Objective: Go beyond correlation to identify true cause-and-effect relationships.

Techniques:

- **A/B Testing:** Randomized experiments to measure treatment effects.
- **Instrumental Variables (IV):** For cases where randomization isn't feasible.
- **Difference-in-Differences (DiD):** Measuring causal effects by comparing trends over time.

Real-World Applications:

- Personalizing customer interventions (e.g., discounts).
- Evaluating the impact of policy changes.
- Optimizing healthcare treatments.

Stability Advantage: Anchoring models on true causal factors improves robustness under changing conditions.

Tools & Libraries (Overview)

- Microsoft Responsible AI Toolbox: https://github.com/microsoft/responsible-ai-toolbox
  - Error Analysis
  - Fairness Assessment Fairlearn
  - Model Interpretability InterpretML
  - Counterfactual Analysis DiCE
  - Causal Analysis: EconML
  - Data Balance Responsible AI helps users gain an overall understanding of their data
- HAX Playbook: Plan for common human-AI interaction failures early: https://microsoft.github.io/HAXPlaybook/
- MAPIE - Model Agnostic Prediction Interval Estimator
- alibi: model inspection and interpretation
- TorchCP: toolbox for conformal prediction research on deep learning models

Books:

- Book and Course: https://github.com/valeman/awesome-conformal-prediction
- Causality for Machine Learning: https://ff13.fastforwardlabs.com/
- Causal Machine Learning Book: https://causalml-book.org/

Organizational Best Practices for Responsible AI

What Does Responsible AI Mean in Organizations?

- Creating a culture of **responsibility**: Everyone in a company, from engineers to executives, takes ownership of AI ethics.
- Encouraging team **collaboration**: Engineers work with ethicists, legal experts, and designers to address ethical concerns together.
- **Monitoring** AI continuously: Always check your AI models for fairness, safety, and unexpected changes after deployment.

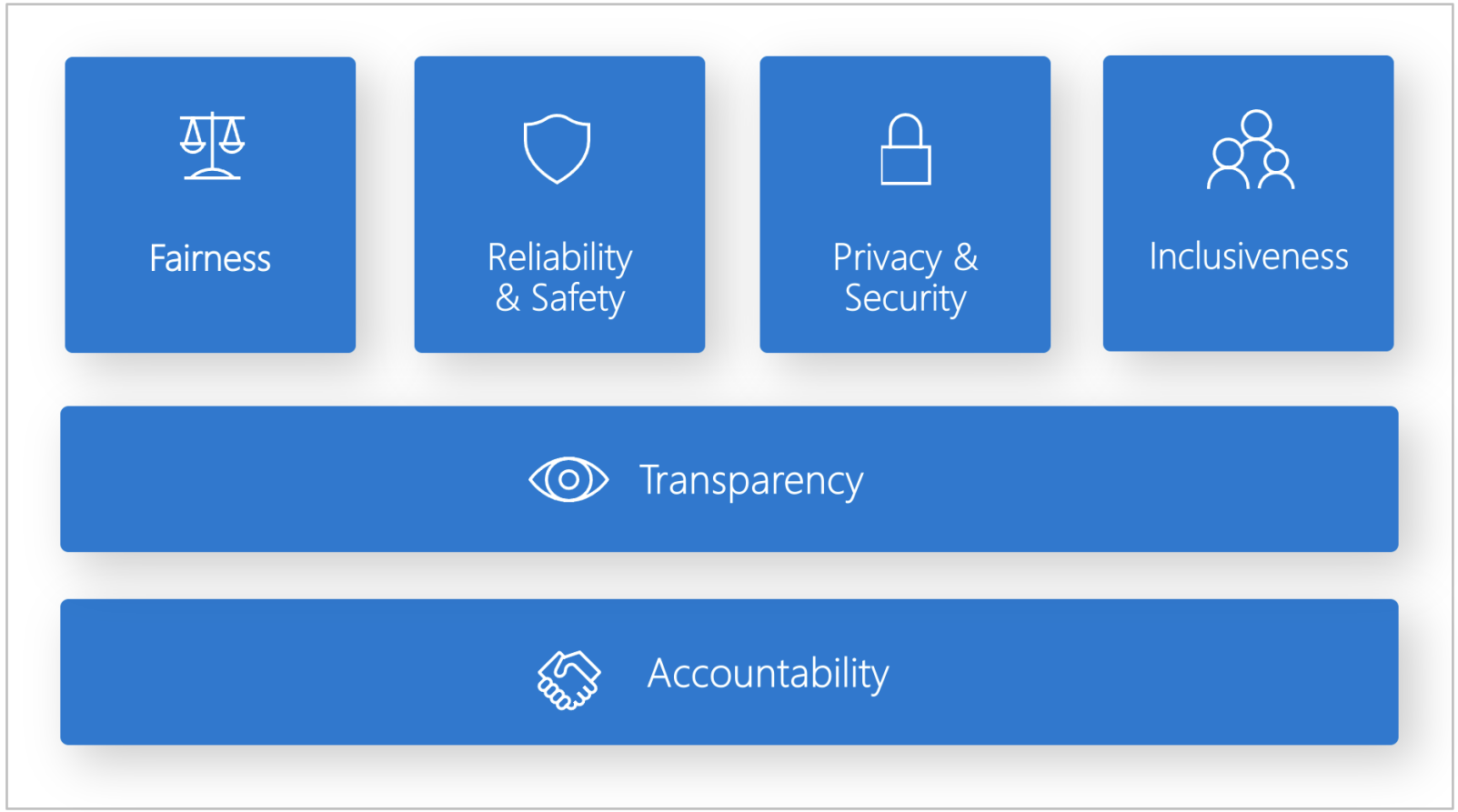
Organizational Best Practices for Responsible AI

Examples from Real Companies:

- **Microsoft Responsible AI Standard:** Microsoft ensures their AI is fair, transparent, and reliable
- **Google AI Principles:** Google makes AI tools that are socially beneficial, like AI for detecting diseases, while avoiding AI misuse.
- **Meta's AI Practices:** Meta builds AI systems that are both fair and privacy-centric, enhancing user interactions with their platforms.
- **Amazon AI Policy:** Amazon prioritizes fairness, safety, and reliability in AI, with rigorous testing for tools like Alexa and warehouse robots.

Responsible AI Standard by Microsoft

LINK



How Can YOU Learn From This?

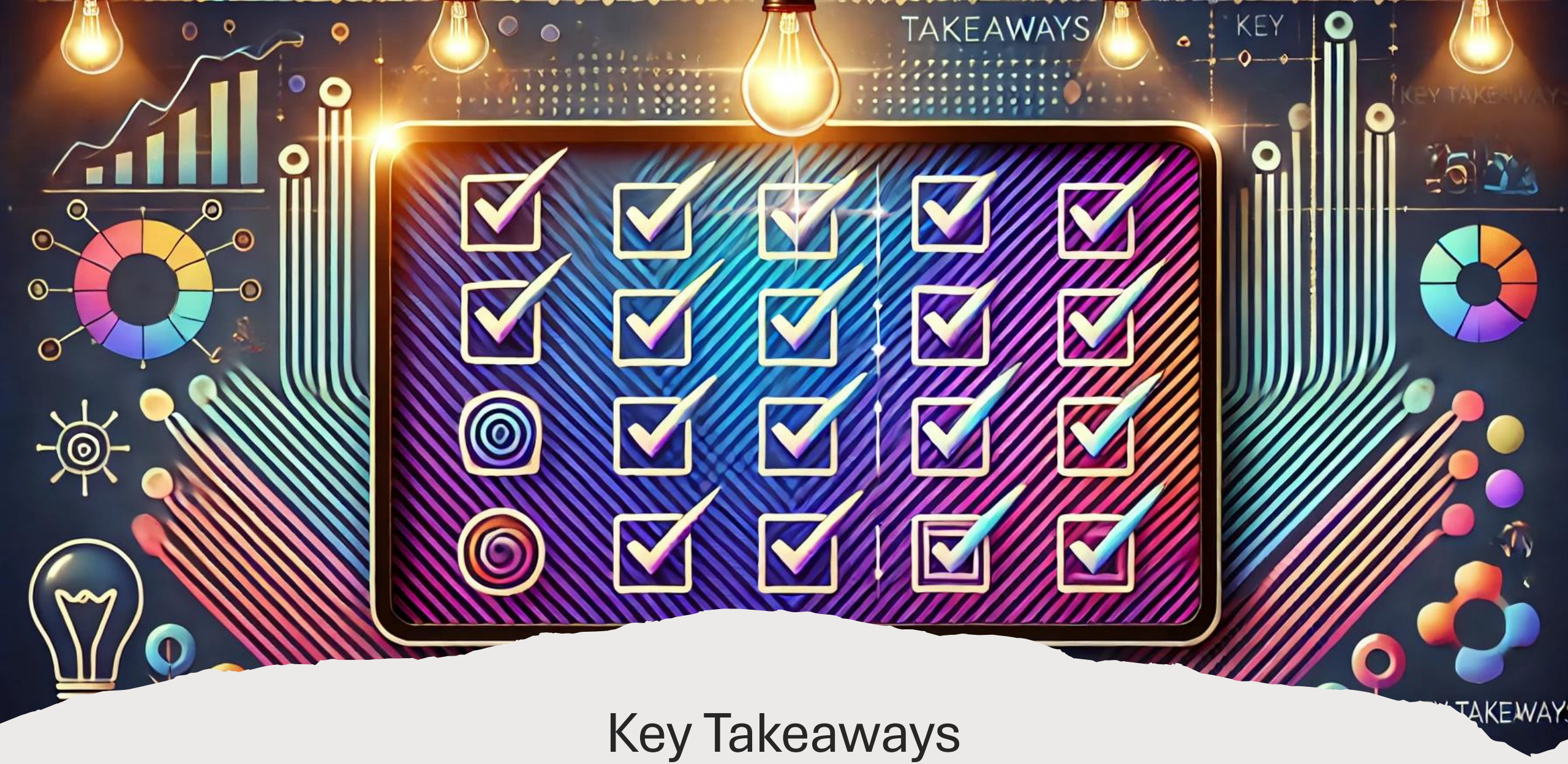
1. **Work as a Team:** Don’t solve AI problems alone—learn to collaborate!
2. **Ask Ethical Questions:** “Who will this impact?” “Is it fair?” “Could it harm someone?”
3. **Use What You Know:** Apply lessons like fairness metrics or model testing to check your AI ideas or validate prototypes.

Stability Meets Responsibility in ML

- **Validate Regularly**: Implement routine data validation to identify distribution shifts or concept drift early.
- **Train Robustly**: Incorporate adversarial examples and simulate synthetic worst-case scenarios for resilience.
- **Monitor Continuously**: Set up real-time monitoring systems with automated alerts for bias, performance, and fairness deviations.
- **Actionable Insights**: Ensure monitoring tools generate actionable steps for immediate mitigation of detected issues.

Future Directions

- **Emergence of Generative AI and LLMs**: Challenges in fairness, interpretability, and unintended consequences with expanding model capabilities.
- **Real-Time Drift Adaptation**: Leveraging adaptive learning techniques, including online and reinforcement learning, for proactive system updates.
- **Strengthening Legal and Ethical Standards**: Anticipating stricter global regulations like the EU AI Act and ethical principles for AI accountability.
- **Human-AI Collaboration**: Fostering symbiotic relationships where AI enhances human decision-making while staying under ethical oversight.



Key Takeaways

Summary & Key Takeaways

- **Core Principles of Responsible ML:** Ethics, transparency, reliability, and accountability are the foundation of trustworthy AI.
- **Bias Mitigation and Fairness:** Apply fairness metrics, adversarial debiasing, and data balancing to ensure equitable outcomes and minimize harm.
- **Interpretability and Explainability:** Use tools like SHAP, LIME, and Model Cards to enhance trust, accountability, and stakeholder understanding.
- **Robustness and Stability:** Design systems resilient to data drift, adversarial attacks, and operational changes through proactive monitoring and testing.
- **Organizational Practices:** Build a culture of responsibility with interdisciplinary collaboration, ongoing monitoring, and robust MLops frameworks.
- **Future-Proofing AI:** Address emerging challenges from large language models, evolving regulatory frameworks, and real-time adaptive systems.

Thank you for your attention